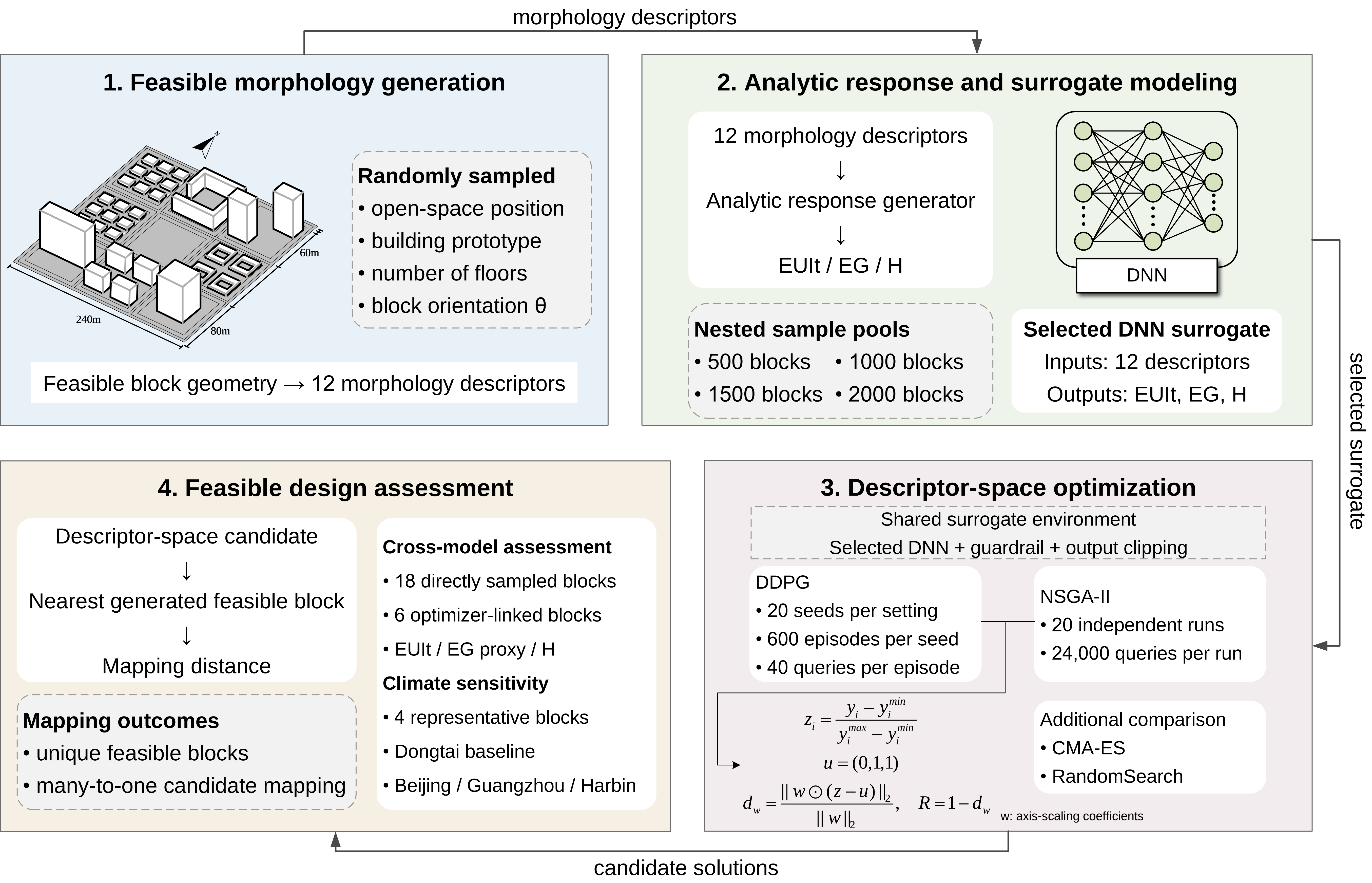
morphology descriptors
1. Feasible morphology generation
60m
240m
80m
Randomly sampled
• open-space position
• building prototype
• number of floors
• block orientation θ
Feasible block geometry → 12 morphology descriptors
2. Analytic response and surrogate modeling
12 morphology descriptors
↓
Analytic response generator
↓
EUIt / EG / H
DNN
Nested sample pools
• 500 blocks • 1000 blocks
• 1500 blocks • 2000 blocks
Selected DNN surrogate
Inputs: 12 descriptors
Outputs: EUIt, EG, H
selected surrogate
4. Feasible design assessment
Descriptor-space candidate
↓
Nearest generated feasible block
↓
Mapping distance
Mapping outcomes
• unique feasible blocks
• many-to-one candidate mapping
Cross-model assessment
• 18 directly sampled blocks
• 6 optimizer-linked blocks
• EUIt / EG proxy / H
Climate sensitivity
• 4 representative blocks
• Dongtai baseline
• Beijing / Guangzhou / Harbin
3. Descriptor-space optimization
Shared surrogate environment
Selected DNN + guardrail + output clipping
DDPG
• 20 seeds per setting
• 600 episodes per seed
• 40 queries per episode
NSGA-II
• 20 independent runs
• 24,000 queries per run
$z_i = \frac{y_i - y_i^{min}}{y_i^{max} - y_i^{min}}$
$u = (0,1,1)$
$d_w = \frac{\| w \odot (z-u) \|_2}{\| w \|_2}, \quad R = 1 - d_w$
w: axis-scaling coefficients
Additional comparison
• CMA-ES
• RandomSearch
candidate solutions